

BIRLA INSTITUTE OF TECHNOLOGY AND SCIENCE, PILANI

FIRST SEMESTER 2023 – 2024

CE F230 Civil Engineering Materials (Comprehensive Examination)

20-12-2023 (Wednesday)

Duration: 180 Mins. (9 am – Noon)

Max. Marks: 52 + 28 = 80

Name:

ID:

Student sign:

Invigilator sign:

PART A (CLOSED BOOK - 52 Marks), Max. time: 2 hours

✎ Write NEAT and CLEAN and to the point answers.

✎ Use the space judiciously. Use Part B answer book (back pages) for any rough work

1. Briefly define the following in context of civil engineering materials [16]:

Tempering:
Quoin closer:
Chisel:
Glass transition temperature:
ASR:
ITZ:
Pozzolan:
Urban heat island:
Nominal maximum aggregate size:
Gradation:
Bulking:
Entrained air:
Superplasticizer:
Drying Shrinkage:
Plastic shrinkage:
Contraction joint:

2. How does creep in concrete lead to loss of prestress [4].

3. Briefly explain any 4 improvements in wood after seasoning. Compare air vs kiln seasoning. [4]

Air	Kiln

4. Derive the formula for flexural strength in 3-point bending test. (Assume: P= max. total load, L= span length, b= specimen width, d= specimen depth) [4]

5. Compare the stress-strain relationship for FRP composite and steel. Mention 2 applications of FRP in construction industry. [4]

6. Clearly explain using suitable figures why compressive strength of a standard concrete cube is observed to be higher than that of a standard concrete cylinder? [4]

7. You have two FRC specimens A & B. The only difference in their mix design being of fiber type. ‘A’ uses asbestos fibers. ‘B’ uses PP fibers. For the two specimens, compare the following properties (a) bond with matrix (b) ductility of the composite (c) resistance of the composite to impact loads (d) tendency for fiber pull out. [4]

8. Differentiate between true stress and engineering stress. Tensile test was conducted on a TMT steel circular rebar to measure its mechanical properties. Given that the percentage elongation observed was 18% and elongation length between gauge points was 11.8 cm. What was the diameter of steel bar used for the test? [4]

9. What is the purpose of [4]:

TFOT:

Flash point test:

Adding sand to lime mortar:

Not using fat lime for thick walls:

10. Draw four types of bonds in brick masonry [4].

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PART B (OPEN BOOK - 28 Marks)

1. Concrete mix design (IS 10262:2019)

The mix proportioning of M50 grade of concrete is to be done for a site of nuclear power plant construction. Due to some unfavorable condition at site, there was a problem with proper storage of cement. Making appropriate assumptions and strictly following the IS code procedure, design the concrete mix based on below mentioned specifications. Providing justifications at steps is must and will carry marks.

1. Stipulations for proportioning:

- a. Type of cement : OPC 53 grade confirming to IS 269 - 2015
- b. Type of mineral admixture : Fly ash confirming to IS 3812 (part-2)
- c. Maximum nominal size of aggregate: 10 mm
- d. Maximum water-cement ratio : refer to IS 456
- e. Minimum-cement content : refer to IS 456
- f. Workability : 110 mm (slump)
- g. Exposure condition : Extreme (for reinforced concrete)
- h. Method of concrete placing : Workmanship
- i. Type of aggregate : Crushed angular aggregate
- j. Maximum cement (OPC) content : 550 kg/m³
- k. Chemical admixture type : Superplasticizer ECMAS HP 890
- l. Based on experience, assume that
 - a. Water content reduction due to SP = 22%
 - b. Volume of SP = 6 liters/m³ of concrete
- m. Based on project requirement, assume that
 - a. Only 1 % defect level is allowed
 - b. Percentage of fly ash used = 15%
 - c. Increase in cementitious material content = 5%

2. Test data for materials

- a. Cement used : OPC 53 grade confirming to IS 269 – 2015
- b. Specific gravity of cement : 3.2
- c. Fly ash : Confirming to IS 3812 (part-2)
- d. Specific gravity of fly ash : 2.2
- e. Chemical admixture : Superplasticizer confirming to IS 9103
- f. Specific gravity of:
 - i. Coarse aggregate : 2.72
 - ii. Fine aggregate : 2.62
- g. Absorption capacity:
 - i. Coarse aggregate : 1.5 %
 - ii. Fine aggregate : 0.5 %
- h. Moisture content of aggregate [As per IS 2386 (Part 3)]
 - i. Coarse aggregate : 0.5 %
 - ii. Fine aggregate : 3 %
- i. Sieve Analysis results for fine aggregate:

IS Sieve size (mm)	10	4.75	2.36	1.18	0.6	0.3	0.15
% Passing	100	95	90	80	70	14	5

***** Best wishes *****